Sepia

How Sepia works, from slide to sign-out

A plain-language guide to the app: what a pathologist sees, what data sits underneath, how the models and the database fit together, and what the evaluation actually shows.

Research prototype. Not for clinical use.

Team 44 · 2026 Ashby Prize in Computational Science Hackathon

October 18, 2026

1 · OVERVIEW

Sepia in one page

Sepia is a worklist for breast pathology. It reads each microscope image with pathology AI models, decides how urgently a pathologist should look at it, and shows why, while the pathologist keeps every decision.

The problem it tackles

In most labs, cases are read in the order they arrive. An invasive breast cancer can sit in the queue behind a stack of benign biopsies. AI tools that try to help with this usually give a single score and stop there: they rarely say *why* a case was flagged, and they rarely say how far the flag should be trusted.

What Sepia does about it

- **It sorts the queue by urgency.** Surgical frozen sections come first, then cases that might be invasive cancer, then cases where the model is unsure, then everything else.
- **It says how confident it is, honestly.** Probabilities are calibrated, and each case gets a *confidence set*: the list of diagnoses that cannot be ruled out. If invasive cancer is anywhere in that list, the case goes up the queue.
- **It shows its reasoning.** A heatmap shows which parts of the image drove the read. Nine named tissue features (for example "tumour infiltrating stroma" or "gland formation") explain the prediction in pathology language. Similar past cases are shown for comparison.
- **It prepares the next steps, but never acts on them.** A rule engine drafts follow-up stain orders. Nothing is ordered until the pathologist approves.
- **It is open about its own results.** A model-evaluation page shows every metric with its uncertainty, including the places where the approach did not work.

Headline results

0.998
AUROC for telling invasive cancer apart from everything else (132 held-out images)

31 / 31
invasive images placed in the top two worklist tiers; none left waiting

90.2%
of confidence sets contained the true diagnosis, against a 90% target

20 → 6
minutes: simulated mean wait before an invasive case is first opened, versus reading in arrival order

These numbers come from a small public dataset and a simulation. They are research results, not clinical validation. Section 7 explains each one and section 8 lists the limits.

Contents 2 How a case flows · 3 The data · 4 Models and pipeline · 5 The database · 6 Inside a case · 7 What the analysis shows · 8 Outcomes and limitations · 9 Credits

2 · THE FLOW

How a case flows through the app

The heavy computing happens once, ahead of time. The app itself never calls a GPU or a language model while someone is reviewing a case. It reads saved results and records what the pathologist decides.

On the Delta supercomputer

Prepare the images, cut them into tiles, run the foundation models to turn each tile into a numeric "embedding", encode the text prompts, and segment cell nuclei.

On the laptop (analysis)

Train the classifiers, calibrate them, compute confidence sets, concept scores, reliability, similar cases and margin reads. Write one JSON file per case, and run the evaluation.

In the app (review)

Load those case files into a local SQLite database, replay case arrivals, sort the worklist, and save every decision the pathologist makes.

A day in the app, step by step

1

Cases arrive

The arrival replay releases precomputed cases over a simulated working day.

2

Queue sorts

Each case gets a tier from 1 to 4 and drops into place in the worklist.

3

Pathologist reviews

Heatmap, features, confidence, similar cases and a cited summary.

4

Decide on orders

Approve or edit the drafted stains, look closer, or dismiss the flag.

5

Sign out and route

The result goes to the oncology inbox, or a margin call goes to the operating room.

How the queue is ordered

Every case is placed in one of four tiers by a single shared rule. The same rule is used in the data pipeline, in the backend and in the workflow simulation, so they cannot drift apart.

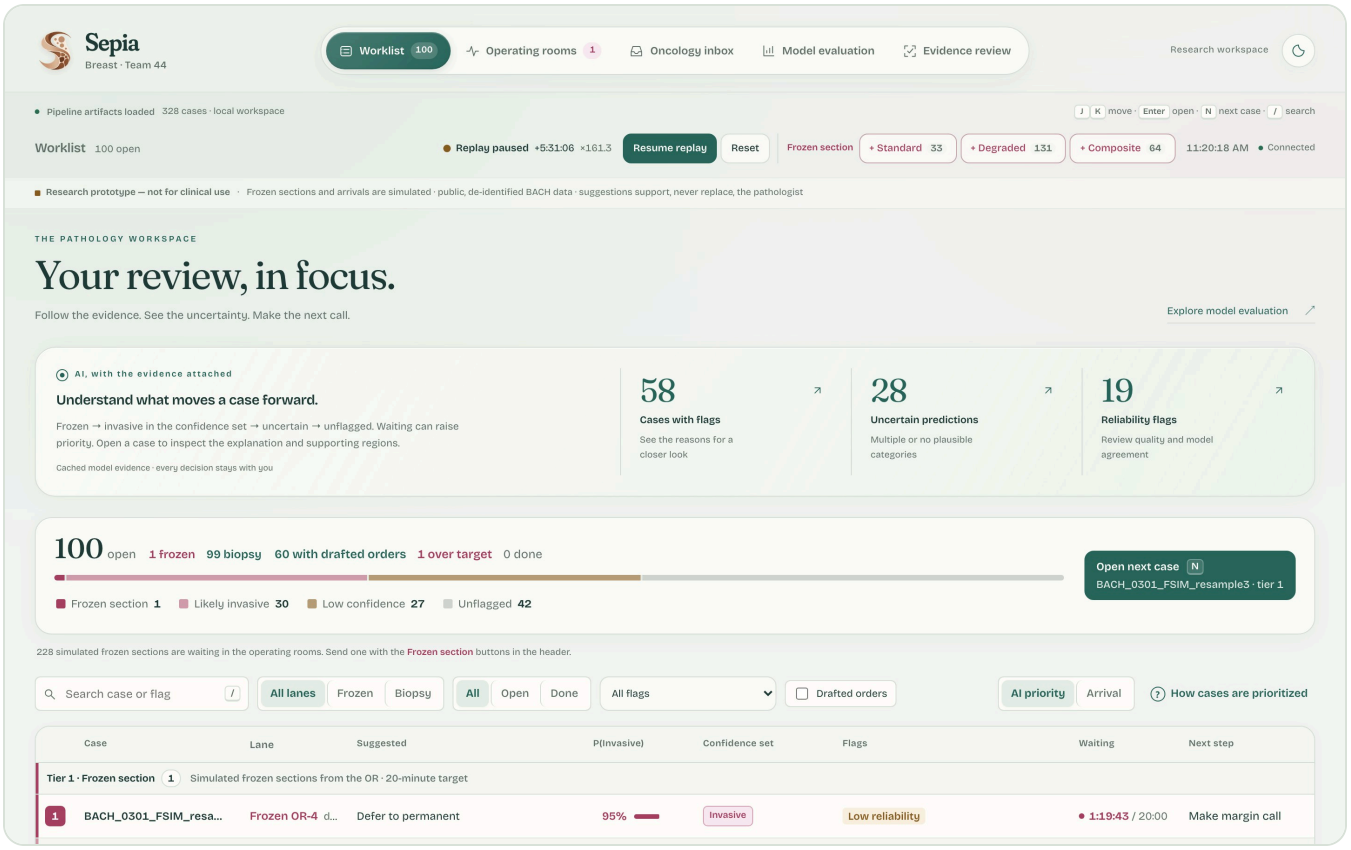
Tier 1 · Frozen section
A simulated surgical specimen. A 20-minute timer is running because a surgeon is waiting.

Tier 2 · Likely invasive
Invasive cancer is the top prediction, *or* it is anywhere in the confidence set. This is the safety rule.

Tier 3 · Low confidence
The confidence set has zero or several diagnoses, the reliability check failed, or too little tissue was readable.

Tier 4 · Unflagged
Everything else. "Routine" does not mean "cleared": a pathologist still reads every case.

An optional aging guard can promote a case that has waited too long. At the configured targets it never triggered within a simulated 8-hour day, so it is not claimed as a benefit.

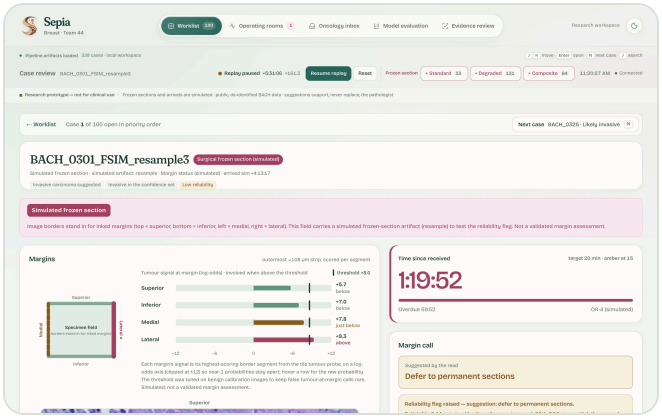


The worklist. The header shows the replay clock and buttons to send in simulated frozen sections (standard, degraded or composite). The summary bar counts open cases by tier. Below, cases are grouped by tier; each row shows the suggested diagnosis, the calibrated P(invasive), the confidence set, any flags, the waiting time against the target, and the next step. A toggle switches between AI priority and plain arrival order.

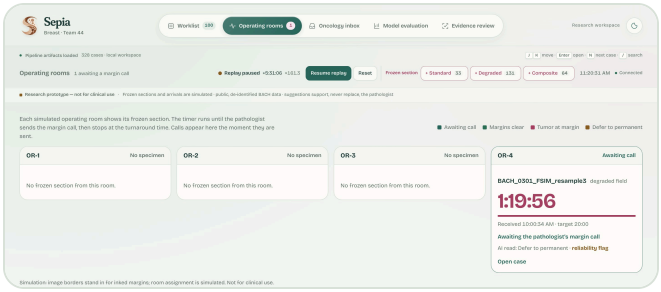
The two lanes: biopsy and surgical

Biopsy lane. The normal case. The pathologist opens it, reviews the evidence, decides on the drafted orders and signs out. Sign-out only becomes available after the order decision has been made, and the signed-out result appears in the **oncology inbox**.

Surgical lane (simulated). This mimics an intra-operative frozen section, where a surgeon waits for the pathologist to say whether tumour reaches the edge of the removed tissue. BACH has no real surgical margins, so Sepia uses the four borders of each image as stand-ins for the inked margins (top = superior, bottom = inferior, left = medial, right = lateral). Some specimens are made deliberately harder: *degraded* versions carry blur, JPEG compression, stain shift or resampling to imitate frozen-section artifacts, and *composite* fields join a benign half to a cancer half so the correct answer is known exactly. The pathologist chooses *margins clear*, names the involved margins, or *defers to permanent sections*. The call stops the timer and appears on the operating-room board.



A simulated frozen section. Each margin gets a tumour score; bars past the threshold line count as involved. Here the reliability check flagged the degraded image, so the suggestion is to defer.



The operating-room board. The surgeon-facing screen: one card per room, the running timer, and the pathologist's call as soon as it is sent.

Frozen sections, margins, operating rooms and arrivals are all simulated. The app connects to no hospital system; approvals and routed results live only in the local database.

What data is used, and how

Every image in the worklist comes from one public, de-identified breast dataset. Other datasets are used only to teach the model what tissue features look like, or to sanity-check those features.

Dataset	What it is	How Sepia uses it
BACH (ICIAR 2018)	400 breast microscopy images (2048 × 1536 px), 100 each of Normal, Benign, In situ carcinoma and Invasive carcinoma, plus 10 annotated whole-slide images.	The dataset of record. Every case in the app is a BACH image. The whole-slide annotations are used to test the simulated margin read.
Quilt-1M	About one million pathology image–caption pairs taken from educational videos.	1,017,708 caption rows scanned; 26,837 breast captions mined to find example images of each tissue feature.
PathGen	Pathology images with generated descriptions.	A second source of feature example images and captions.
NCT-CRC-HE-100K	Colorectal tissue tiles in nine tissue classes.	About 300 tiles per class, used only as a cross-organ check that features like "necrosis" or "lymphocytes" behave sensibly. Not used for breast validation.

How the BACH images are split

The images follow the published **THUNDER** benchmark split: **218 training**, **50 calibration** and **132 test** images. Patient IDs are known for 145 images, and the split is checked so that no patient appears in two parts. Images without a patient ID are treated as their own "pseudo-patient" groups. All confidence intervals in this document are bootstrapped over these patient groups, which keeps them honest about how little independent data there is.

- **Training** is used to fit models and to choose between them by cross-validation.
- **Calibration** is used only to set thresholds: the conformal threshold, the reliability flag threshold and the margin threshold.
- **Test** is never used for any choice. It feeds the worklist and the reported results. (One exception, the switch of conformal method, is disclosed in section 7.)

From image to numbers

1. **Tiling.** Each image is cut into a 3 × 4 grid of 512-pixel regions (plus a finer grid of 256-pixel tiles for some steps).
2. **Tissue quality control.** Tiles that are mostly glass or background are dropped. A case needs at least 3 readable regions, otherwise it is flagged "insufficient tissue". A bug found on October 18 had been rejecting pale tissue; fixing it raised the tile pass rate from 75% to 99% (see figure 7.27).
3. **Embedding.** Each tile, plus rotated and flipped copies, is passed through foundation models on Delta. Each model turns the tile into a long vector of numbers that summarises what it sees. The copies are averaged ("test-time augmentation").

4. **Pooling.** Tile vectors are averaged into one vector per image, which the classifier reads.

What the app receives: the case file

The pipeline ends by writing one JSON file per case into `artifacts/cases/`. There are 328 of them: the 132 test images on the biopsy lane, plus simulated surgical variants (standard, degraded and composite). Each file contains everything the screen needs, and nothing is recomputed in the app:

Part of the file	What it holds
`prediction`	Model ID, calibrated probabilities for the four classes, the invasive risk score and the conformal set.
`quality`	Tiles that passed QC, how unusual the image is (out-of-distribution percentile), and the reliability score and flag.
`tiles`	Per-region invasive probability and per-region scores for the nine features. These drive the heatmap.
`concepts`	The nine named features: value, contribution to the read, direction and the regions that score highest.
`similar_cases`, `contrast_case`	The 5 most similar training images, and the nearest image with a *different* diagnosis.
`cells`	Nucleus counts and shape measurements from cell segmentation, when available.
`margin`	Surgical lane only: tumour score per margin, the threshold and the suggested call.
`flag`, `agent`	The tier and the reasons for it, plus the drafted follow-up orders and the rules that produced them.
`findings`, `summary`	Structured findings (F1, F2 …), and a plain-language summary in which every sentence cites the findings it relies on.

The models and the pipeline

Sepia does not train a deep network from scratch. It reuses large pathology "foundation models" as feature extractors and adds small, inspectable models on top.

The foundation models

Model	Role in Sepia
UNI2-h (Mahmood Lab)	The **production read**. A simple logistic-regression "linear probe" on its embeddings makes the four-class prediction.
CONCH (Mahmood Lab)	A vision-language model, so it can compare images with text. It scores the nine named features and powers similar-case search.
MUSK (Stanford)	A second, independent vision-language model. Its own probe gives a "second opinion" that is shown in each case.
H-optimus-0, PathGen-CLIP	Candidates for model selection, and a zero-shot comparison.
HoVer-Net (PanNuke)	Finds and classifies cell nuclei for the optional cell measurements: 108,661 nuclei on 1,070 tiles.
gemma-4-31b-it (NCSA Lumen)	Writes the short case summary *offline* from structured findings only. Every number and claim is checked against those findings.

How the production model was chosen

Eight candidates were compared by cross-validation inside the training split, grouped by patient: linear probes on each encoder, an attention-based model (ABMIL), a fusion of the top three, and three "concept bottleneck" models that predict only from the named features or cell measurements. The rule was set in advance: take the lowest cross-validated error, then pick the simplest candidate within one standard error of it. That picked the **UNI2-h linear probe**. The test set was not used for this choice.

Turning scores into trustworthy confidence

- **Temperature scaling.** Raw model probabilities are often over- or under-confident. One number (T = 0.50), fitted on training predictions, rescales them so that "80% confident" means right about 80% of the time.
- **Conformal prediction sets.** Instead of one answer, the model returns the set of diagnoses it cannot exclude at a 90% target. Thresholds come from the 50 calibration images. One answer in the set means a confident read; two or more means a real differential; the safety rule acts on whether invasive cancer is in the set at all.
- **Reliability score.** Three signals are combined to estimate whether a read is likely wrong: how unusual the image is compared to training tissue, how big the conformal set is, and whether the CONCH and MUSK probes disagree. A low score raises the reliability flag.

Explaining the read

- **Region heatmap.** A tile-level probe gives each region its own P(invasive), so the reader can see where the signal comes from.
- **Nine named features.** Each feature (gland/tubule formation, nuclear pleomorphism, mitoses, tumour infiltrating stroma, cells confined within ducts, normal lobules, benign stroma or fat, lymphocytes, necrosis) is scored two ways: by comparing the image to pathologist-style text descriptions in CONCH, and by a "concept vector" learned from Quilt-1M and PathGen example images. These are *features associated with* each class, not causes and not a Nottingham grade.
- **Similar cases.** The nearest training images by CONCH embedding, with their diagnoses, and a contrast case with a different diagnosis.
- **Drafted orders.** A short rule file maps the confidence set to standard work-ups, for example ER/PR/HER2/Ki-67 when invasive is the top call, or p63/calponin when both in situ and invasive remain possible. These rules are illustrative and need clinical review.

The pipeline, step by step

Step	Runs on	What it does
00–01	Delta CPU	Inventory the datasets, build the split, tile every image, run tissue QC, write display images.
02–03	Delta GPU	Embed all tiles with each foundation model; encode 1,177 feature and class prompts.
04	Laptop (+GPU for cells)	Learn concept vectors, score the nine features, segment nuclei and compute cell features.
05	Laptop	Train and select risk models, temperature-scale, build conformal sets, tile probes and the margin probe.
06–07	Laptop	Reliability score and flag; similar-case retrieval.
08	Laptop	Assemble one JSON file per case, including drafted orders and the cited summary.
eval	Laptop	Write the evaluation files (`artifacts/eval/*.json`) that the Model evaluation page reads.

The database

Model results are read-only files. Everything people *do* in the app goes into one local SQLite database, `artifacts/pathqueue.sqlite`, so that refreshing the browser never loses work.

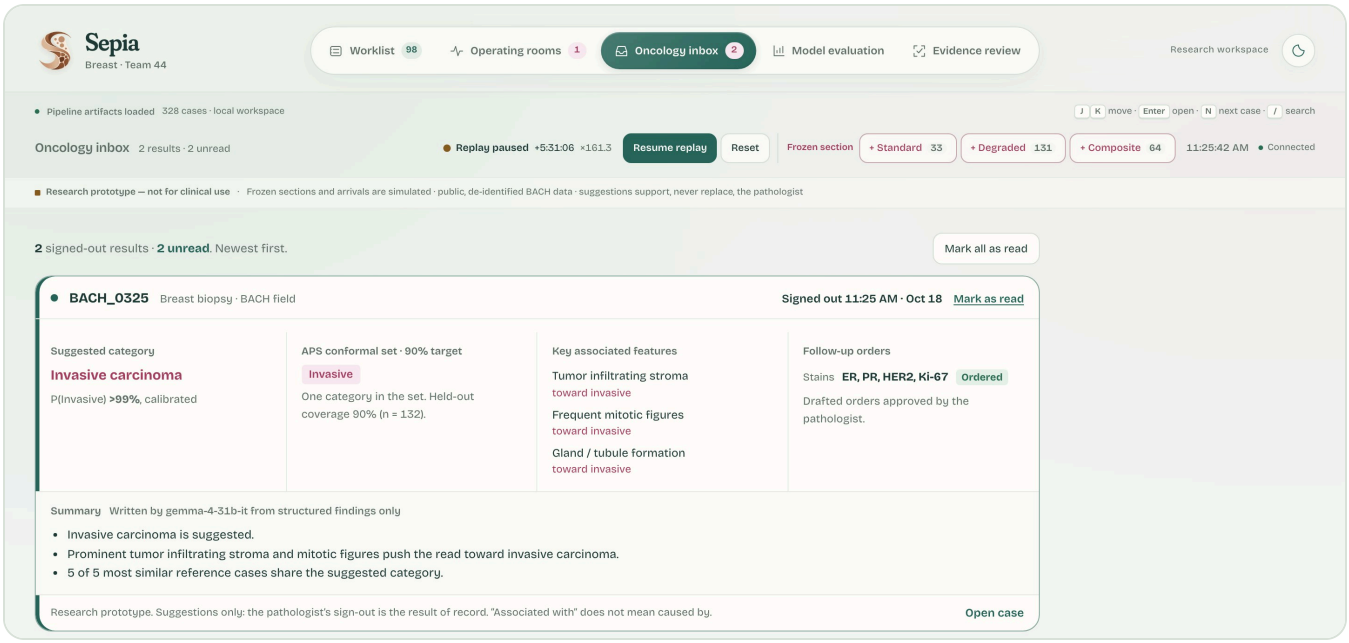
The backend is a small FastAPI server. On start-up it validates every case file and copies it into the database. From then on, the server is the single source of truth: the browser asks it for the worklist, the case and the allowed actions, and posts decisions back. A case's state can only move along allowed transitions, for example sign-out is refused until the order decision has been made.

Table	What a row is	Why it exists
`cases`	One case: its lane, simulated arrival time, the full case file, whether it has arrived, timers.	Holds the replay state: which cases are in the queue and since when.
`orders`	One drafted follow-up order: type, the rule that drafted it, its rationale, the drafted items and the current items.	Keeps the original draft next to any edits, so changes stay visible.
`decisions`	One pathologist action: approve, edit, look closer, dismiss flag, sign out, margin call, with the state before and after.	An audit trail of who decided what, and when.
`routed_results`	One result sent onward, to the surgeon or the oncologist.	Feeds the operating-room board and the oncology inbox.
`events`	Every state change, with wall-clock and simulated time.	A full timeline of each case.
`workspace_state`	Small key–value settings, such as the replay clock.	Lets the replay resume exactly where it stopped.

How the pieces talk

`GET /api/worklist` returns the sorted queue. `GET /api/cases/{id}` returns one case with its current tier, timer and allowed actions. `POST /api/cases/{id}/decision` records an action. `POST /api/sim/…` starts, pauses or resets the replay and injects frozen sections. `GET /api/metrics` serves the evaluation files. `GET /api/health` reports how many cases loaded, which were rejected, and whether any referenced image is missing.

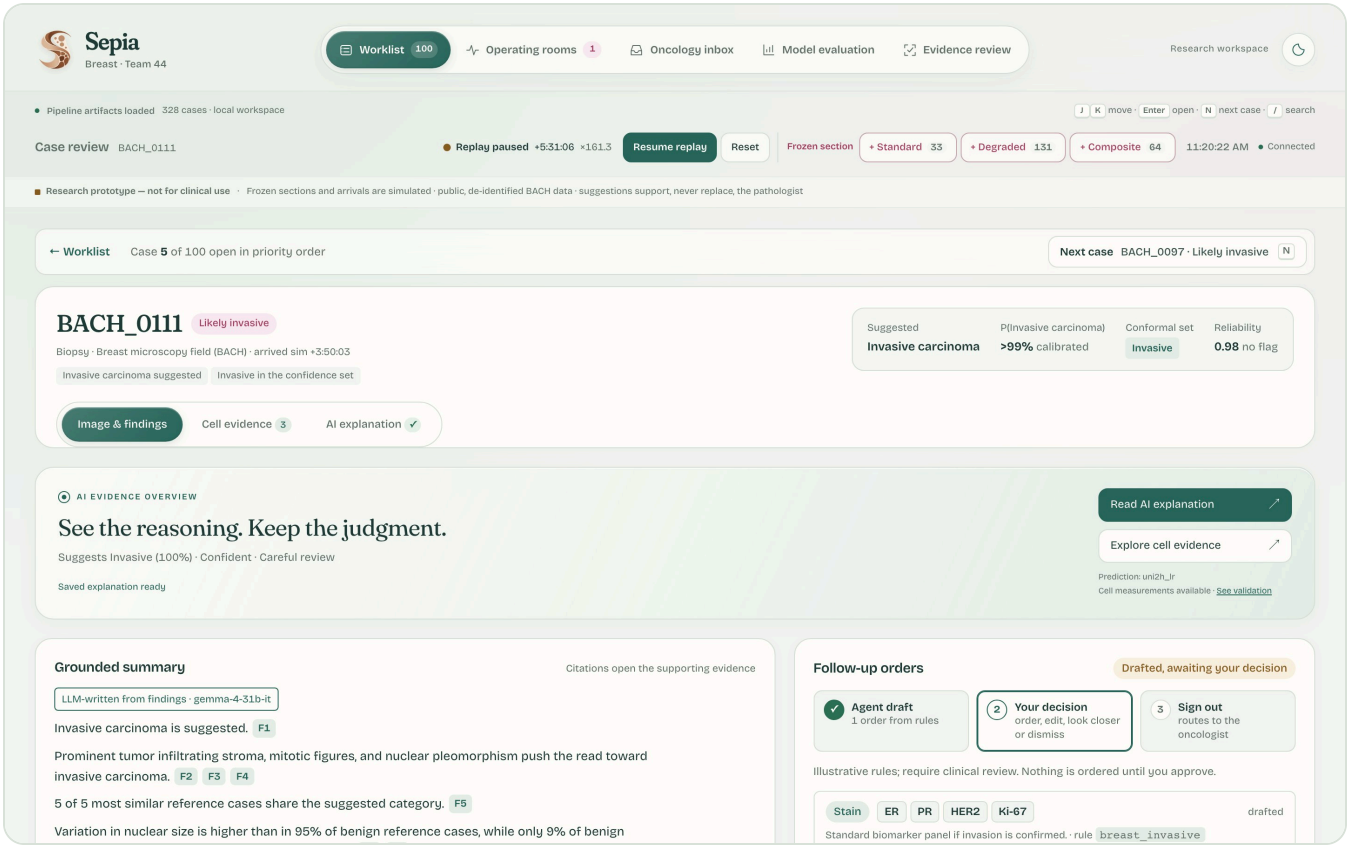
Reset archives the current workspace and starts a fresh replay. It does not change any model output. The app also refuses to fall back silently to synthetic data: if the real case folder is missing, it says so.



The oncology inbox. Signed-out results land here from the `routed_results` table: the suggested category, the confidence set, the key features, the orders the pathologist approved, and the cited summary. (For this screenshot two cases were signed out in a scratch copy of the database.)

Inside a case: the review screen

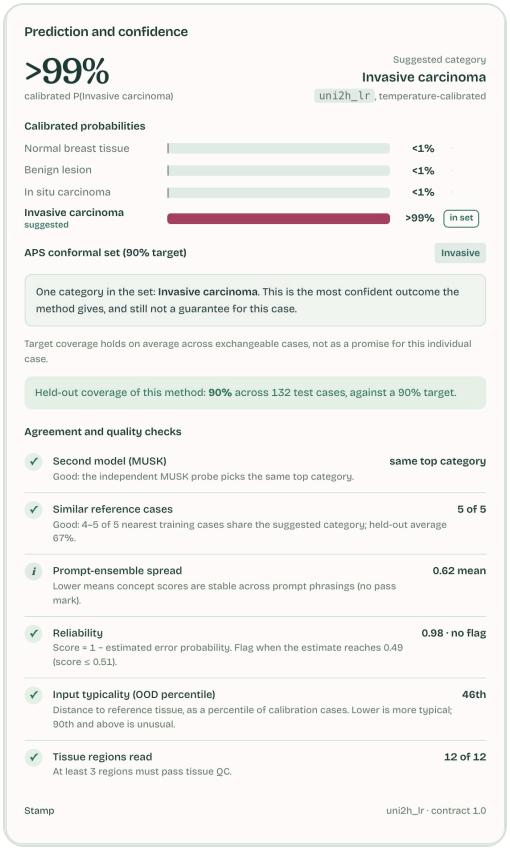
Opening a biopsy shows the read, the evidence behind it, and the decision steps. The example below is BACH_0111, a test image that Sepia suggests is invasive carcinoma.



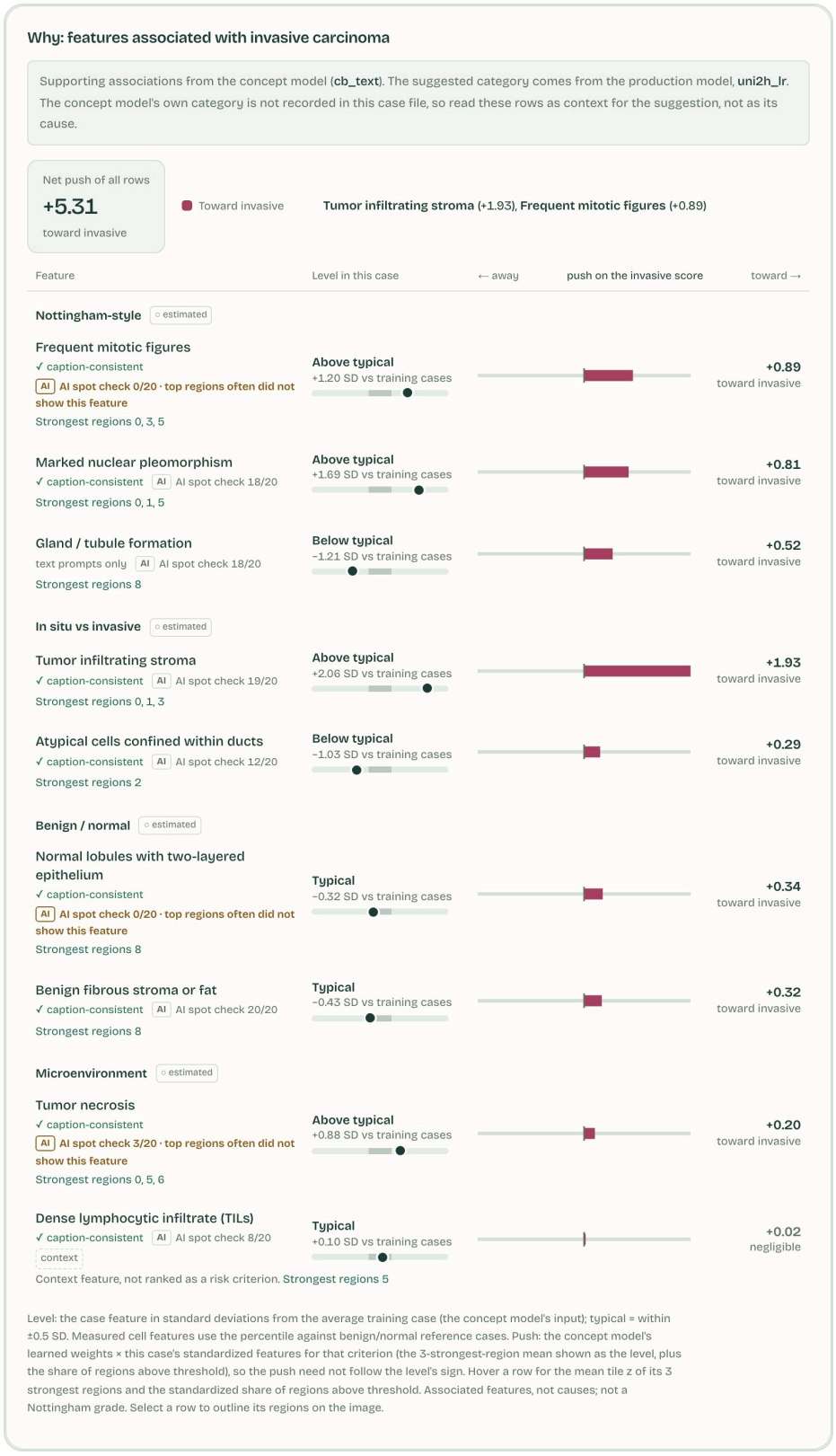
Top of the case view. The header shows the suggestion, the calibrated P(invasive), the conformal set and the reliability score. Below are the grounded summary, where each sentence carries citations (F1, F2 …) that open the supporting evidence, and the three-step order panel: agent draft, your decision, sign-out.



Image evidence. The 3 × 4 regions, shaded by each region's own P(invasive). The layer menu switches the overlay to any of the nine features. Regions that failed tissue QC are hatched.



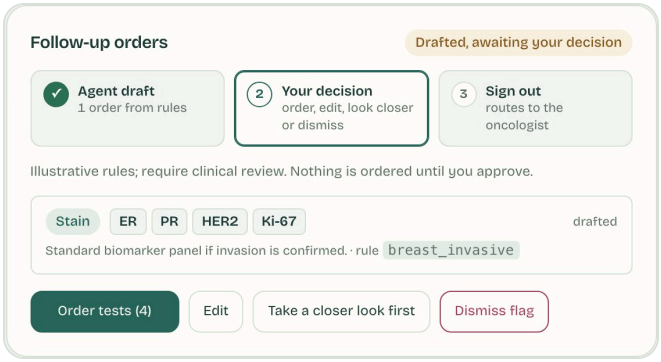
Prediction and confidence. Calibrated probabilities, the conformal set with a plain reading of what it means, and five checks: MUSK agreement, similar cases, prompt stability, reliability and input typicality.



Why: features associated with invasive carcinoma. The nine features are grouped as Nottingham-style, in situ versus invasive, benign/normal, and microenvironment. Each bar shows how much the feature pushed the read and in which direction; clicking one outlines the regions where it scores highest. Labelled "estimated" because these are model associations, not graded components.



Reference comparison. The most similar training images and their diagnoses, plus a contrast case with a different diagnosis.



Follow-up orders. The agent drafts the ER/PR/HER2/Ki-67 panel from rule `breast_invasive`. Nothing is ordered until the pathologist approves or edits.

About the "AI explanation" tab

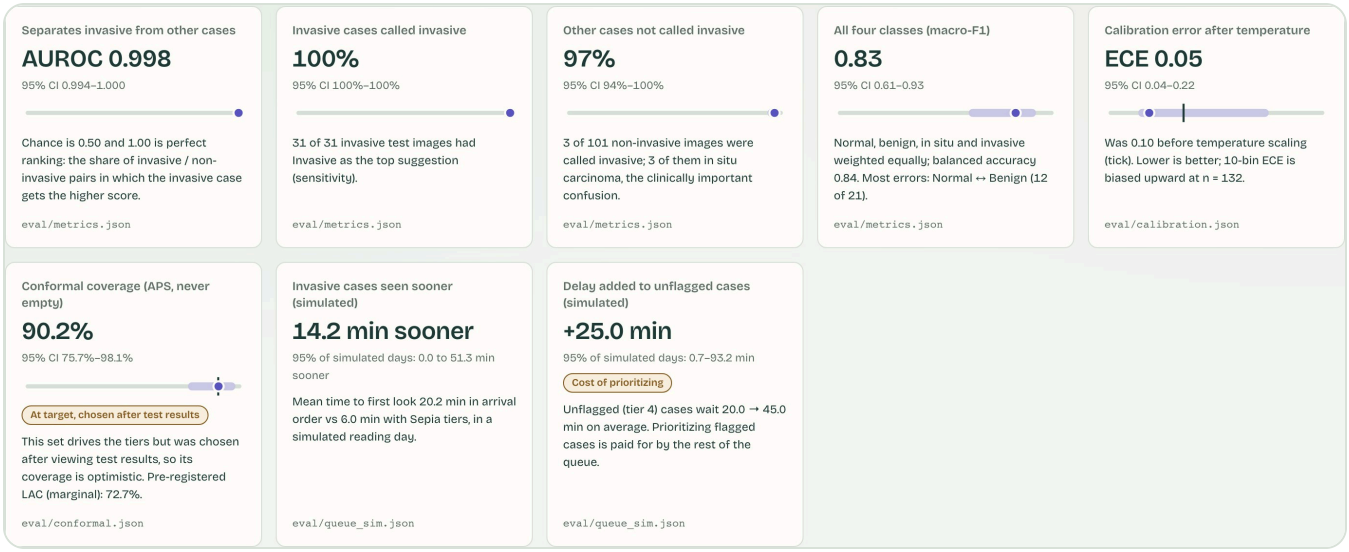
A language model can turn the measured evidence into a readable paragraph. That paragraph adds no new evidence. The proof that the explanations mean something comes from the deletion test, calibration and coverage results in section 7, not from fluent text.

7 · ANALYSIS

What the analysis shows

The Model evaluation page in the app answers one question per chart. Every number is read from the evaluation files at page load and comes with a 95% confidence interval, bootstrapped over patient groups. The charts below are captured directly from that page.

All results are on the **132 held-out test images** (24 patient or pseudo-patient groups) unless a chart says otherwise. With so few independent patients, the intervals are wide: treat the point estimates as encouraging, not definitive.



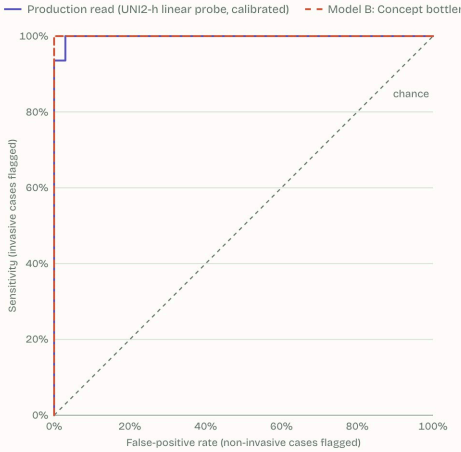
The summary row. The page opens with the headline numbers, each with a status mark and its interval.

7.1 Classification: does it get the diagnosis right?

The most important job is spotting invasive cancer, and here the model is very strong: it ranks invasive images above all others almost perfectly, catches all 31 invasive test images, and wrongly calls only about 3% of the others invasive. Telling all four categories apart is harder. Normal and benign tissue get confused with each other, and in situ carcinoma sits between benign and invasive.

How well does the risk score separate invasive cases?

What this shows The production read reaches AUROC 0.998 (95% CI 0.994–1.000); the explainable concept-bottleneck model B reaches 1.000. The paired difference A − B is −0.002 (95% CI −0.006 to 0.000), so no difference is detectable on this test set. A curve hugging the top-left corner flags invasive cases without flagging others.

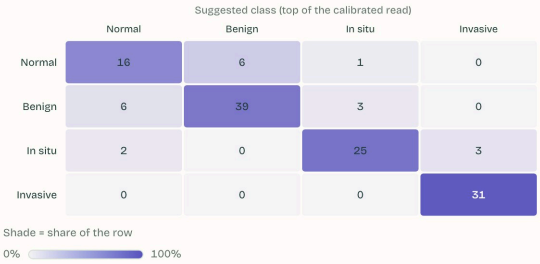


Population: held-out test split, 31 invasive vs 101 other images
Source: `eval/analytics.json`, `eval/metrics.json`

7.1a ROC curve. A curve hugging the top-left corner means invasive cases are flagged without flagging others. AUROC 0.998.

Where do the errors fall?

What this shows Normal 16/23, Benign 39/48, In situ 25/30, Invasive 31/31 correct. 3 of 30 in situ images were called invasive and 0 of 31 invasive images were called in situ. The most frequent error is Normal → Benign (6).

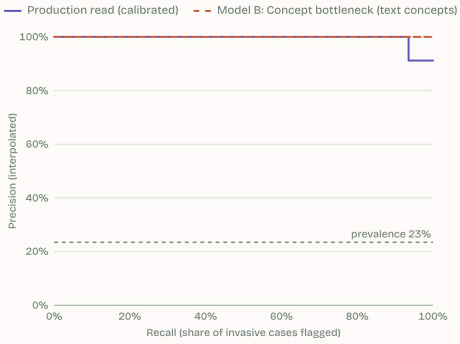


Population: held-out test split, 132 images (rows = true class) Source: `eval/analytics.json`

7.1b Confusion matrix. Rows are the true class, columns the suggestion. Every invasive image is correct; the most common error is Normal called Benign (6 times). 3 of 30 in situ images were called invasive.

When every invasive case is flagged, how many flags are right?

What this shows Average precision 0.994 (model B 1.000). Flagging all 31 invasive images, 91% of the flagged images are invasive at best. The dashed line is the 23% share of invasive images: a useless score would sit on it.



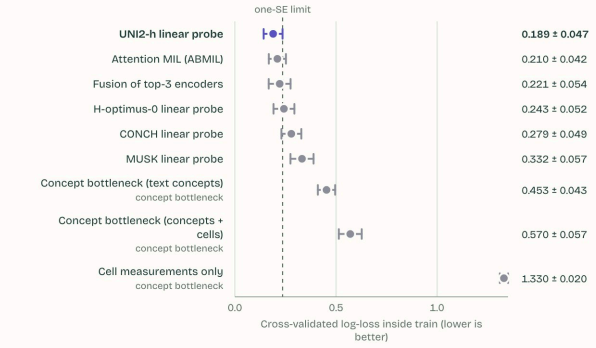
- The curve shows the best precision reachable at each recall; average precision is computed from the raw curve (scikit-learn).

Population: held-out test split, 31 invasive of 132 images Source: `eval/analytics.json`

7.1c Precision–recall. When every invasive image is flagged, about 91% of flags are correct (average precision 0.994).

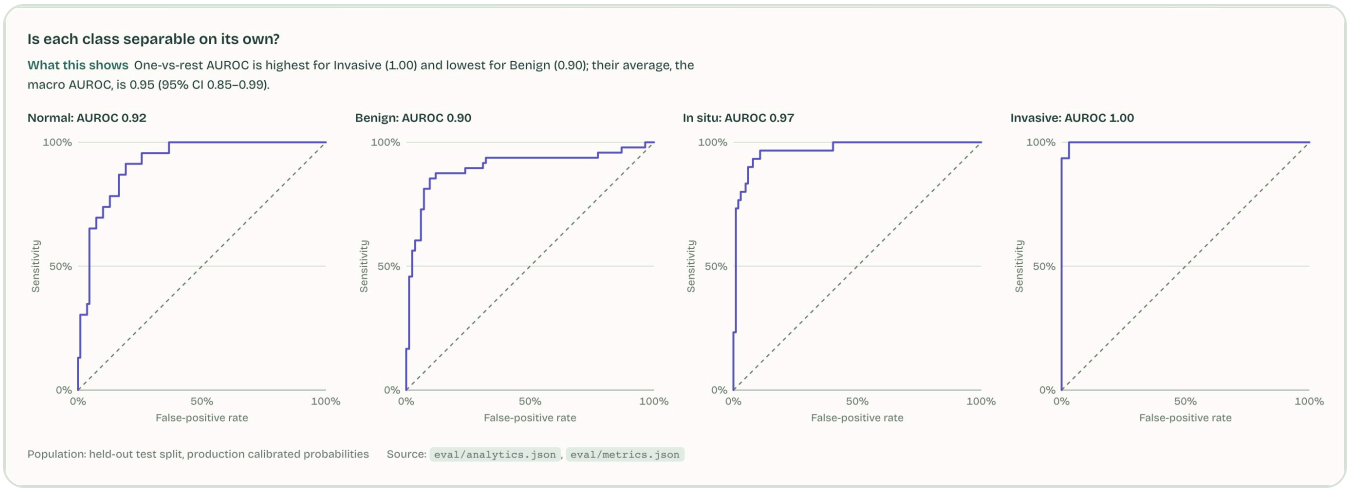
Why this model? Selection inside the training split

What this shows The lowest cross-validated log-loss was UNI2-h linear probe (0.189); the one-SE rule picks the simplest candidate within one standard error (limit 0.236), which is UNI2-h linear probe. The test split was not used (recorded).

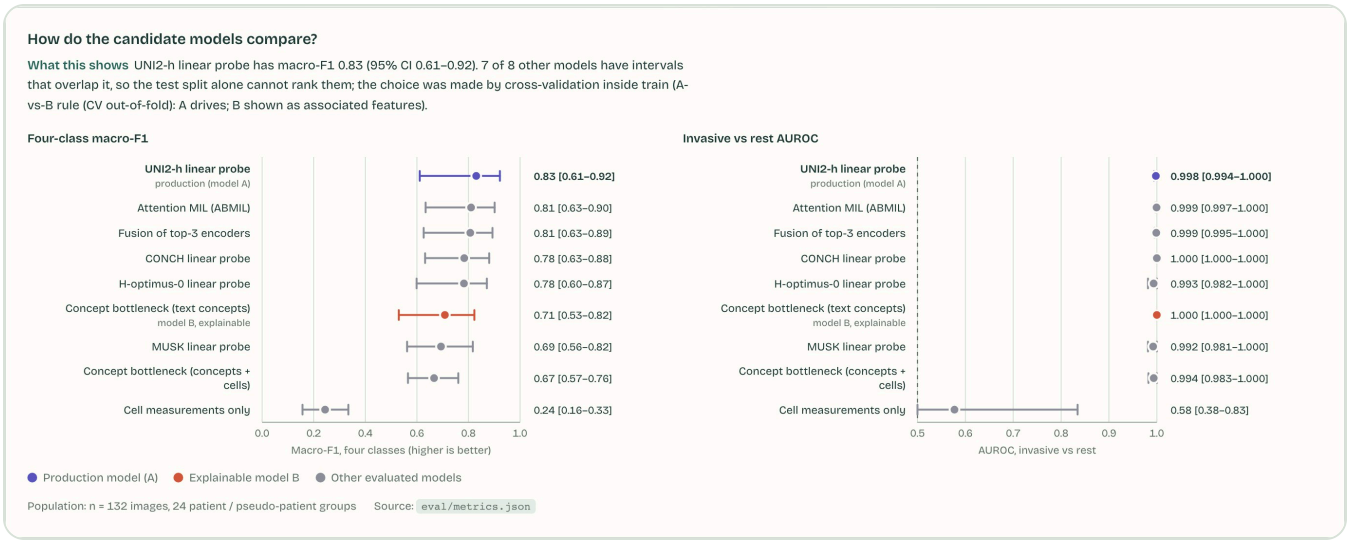


Population: 218 training images, 49 groups, 5-fold grouped CV × 3 repeats Source: `eval/model_choice.json`

7.1d Model selection. Cross-validated error inside training only. UNI2-h has the lowest log-loss and is also the simplest choice within one standard error. Models using cell measurements alone were far weaker.



7.1e One class against the rest. Invasive is the easiest class to separate (AUROC 1.00) and benign the hardest (0.90). The macro average is 0.95.

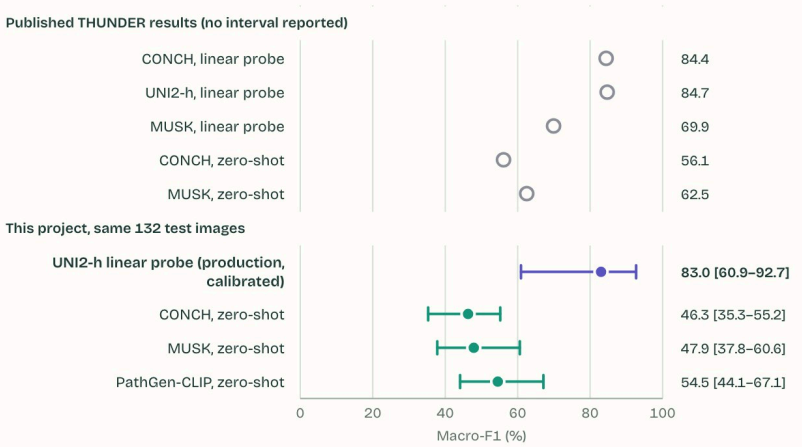


7.1f All candidate models on the test set. Most intervals overlap the production model's, so the test set alone cannot rank them. That is why the choice was made by cross-validation beforehand.

How does this compare with published benchmarks?

What this shows Our production macro-F1 is 83.0% (95% CI 60.9–92.7%); the best published linear probe (UNI2-h) is 84.7%, which lies inside our interval. Published zero-shot text matching scores 7.4–28.3 points below the linear probe of the same encoder; every zero-shot read of ours is also below the production probe.

Same split, different protocol: published: whole-image resize; ours: tiles + TTA. Not compared with the ICIAR 2018 challenge, which used a hidden test set.



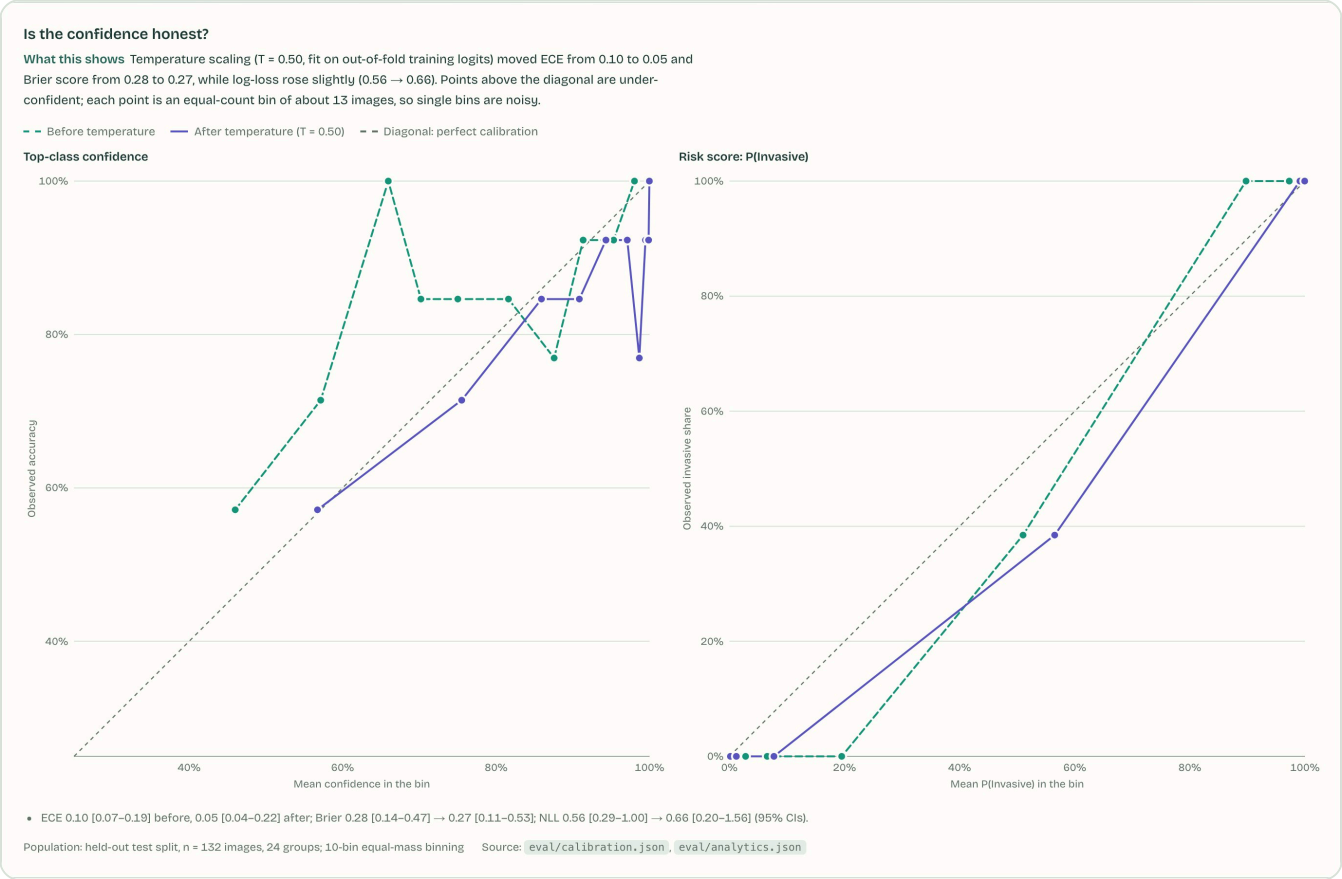
- Published values: THUNDER official leaderboards, BACH F1 scores; verified 2026-10-17 at https://mics-lab.github.io/thunder/leaderboards/.

Population: 132 test images, 24 groups (ours); published values carry no CI Source: `eval/thunder.json`

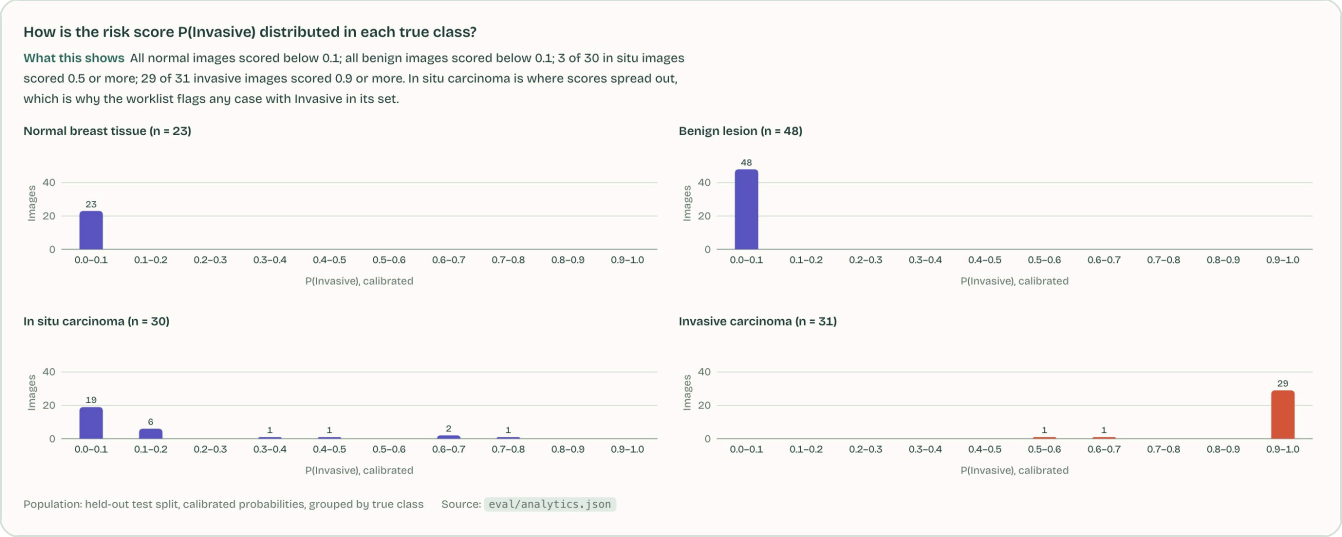
7.1g Published benchmark. On the same THUNDER split, Sepia's macro-F1 (83.0%) matches the best published linear probe (UNI2-h, 84.7%), which lies inside our interval. The protocols differ (we use tiles and augmentation; the benchmark resizes the whole image), so this is an anchor, not a head-to-head win. Zero-shot text matching, with no training at all, scores well below.

7.2 Confidence: can the numbers be trusted?

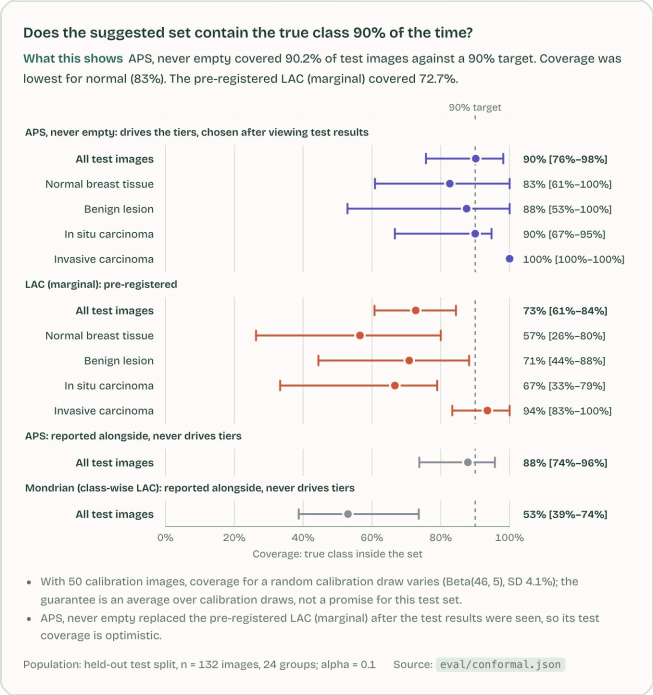
A score is only useful for triage if "90% sure" really means right 9 times out of 10. Temperature scaling halved the calibration error (ECE 0.10 to 0.05). The conformal sets then hit their target: the true diagnosis was inside the set for 90.2% of test images, against a 90% goal.



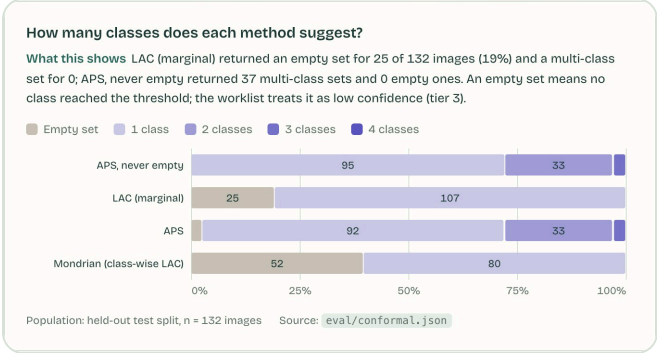
7.2a Reliability diagrams. Points on the diagonal mean stated confidence matches observed accuracy. After temperature scaling (purple) the points sit closer to the line. Each point is a bin of about 13 images, so single points are noisy.



7.2b Where the risk score lands for each true class. Normal and benign images all score below 0.1; 29 of 31 invasive images score above 0.9. In situ carcinoma is where scores spread out, which is exactly where the conformal set and the drafted p63/calponin stains matter.



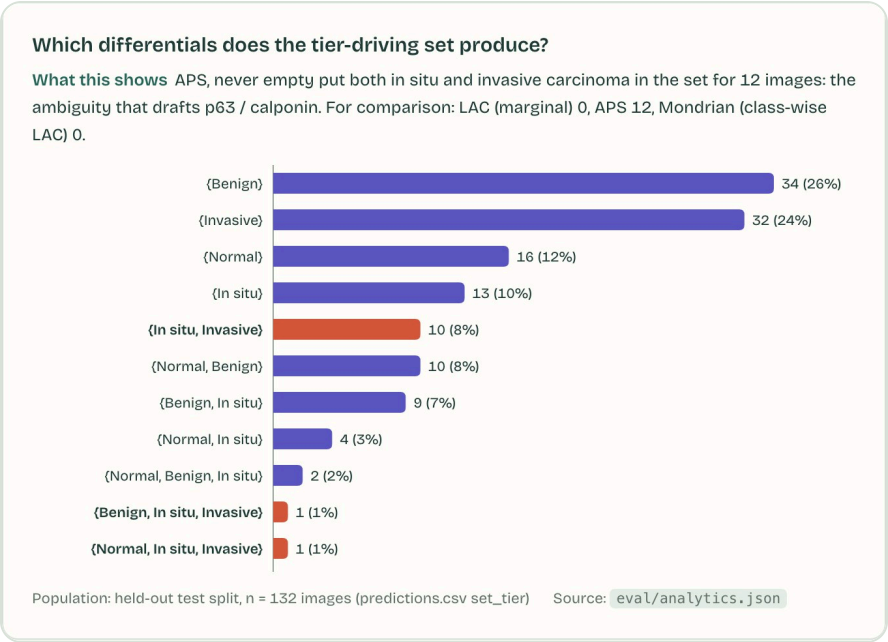
7.2c Coverage by class and method. APS ("never empty") covers 90.2% overall and 100% of invasive images. The pre-registered LAC method only reached 72.7%.



7.2d Set sizes. LAC gave an empty set for 19% of images and never a differential; APS gave 37 multi-class sets and no empty ones.

A disclosed, after-the-fact change

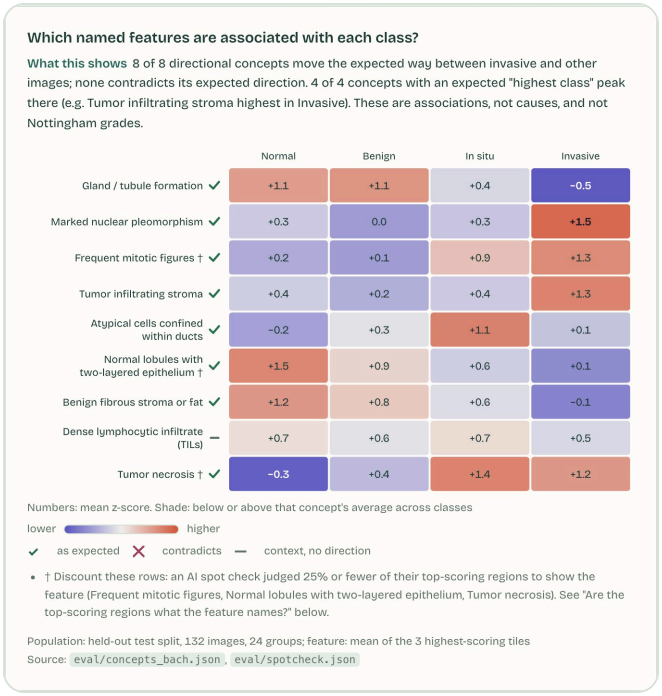
LAC was the method chosen in advance. On the test set it returned empty sets for 19% of cases and gave the reader no differential, likely because the 50 calibration images were easier than the test images. Tiers were switched to APS *after* seeing test results, so APS's 90.2% coverage is optimistic and is not an independent confirmation. Both methods are reported.



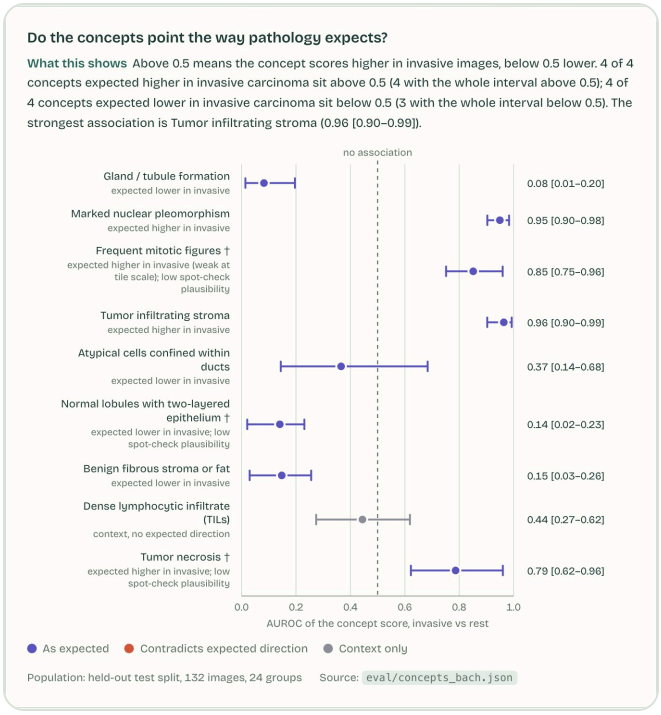
7.2e Which differentials the set produces. For 12 images, both in situ and invasive carcinoma stayed in the set. That is the ambiguity that drafts the p63/calponin order.

7.3 Explanations: do the named features mean anything?

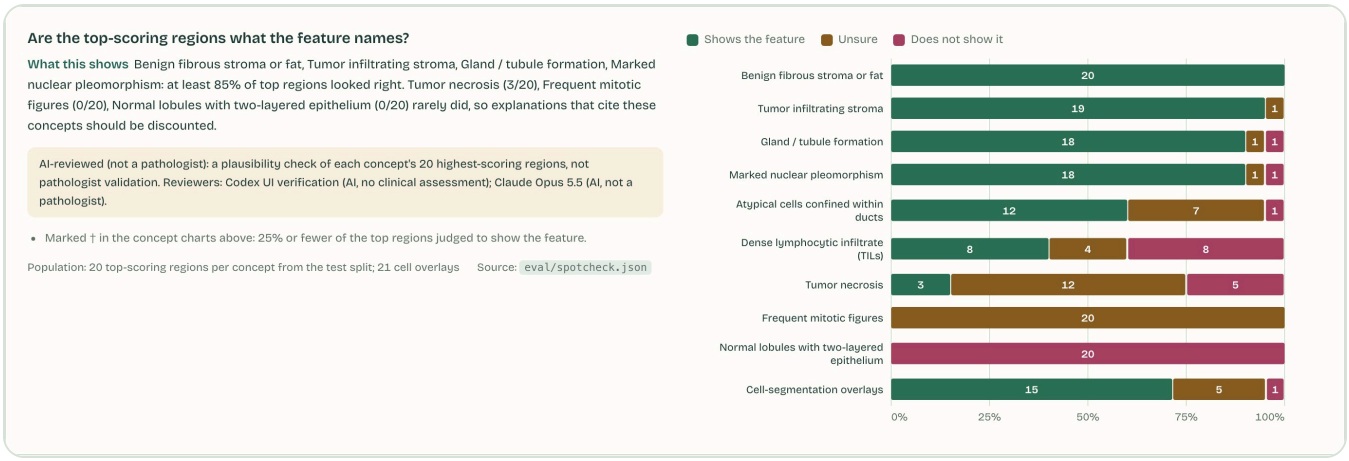
The nine features move in the directions pathology expects: tumour infiltrating stroma, pleomorphism and mitoses rise in invasive images; gland formation, normal lobules and benign stroma fall. But direction alone is not proof. An AI spot check of the highest-scoring regions found that some features point at the right tissue (stroma/fat, invasion, tubules, pleomorphism: 18–20 of 20 looked right) while others do not (mitoses and normal lobules: 0 of 20). Those weak features are labelled as such in the app.



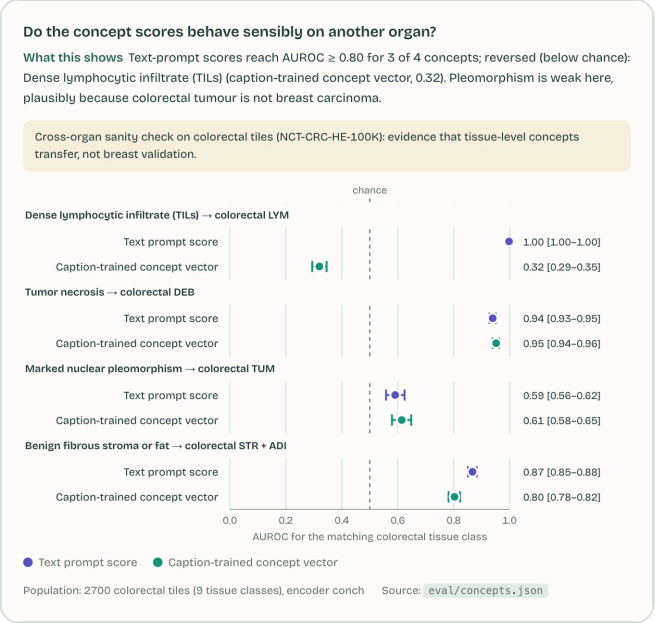
7.3a Mean feature score per true class. Red is above average, blue below. All 8 directional features move the expected way between invasive and other images.



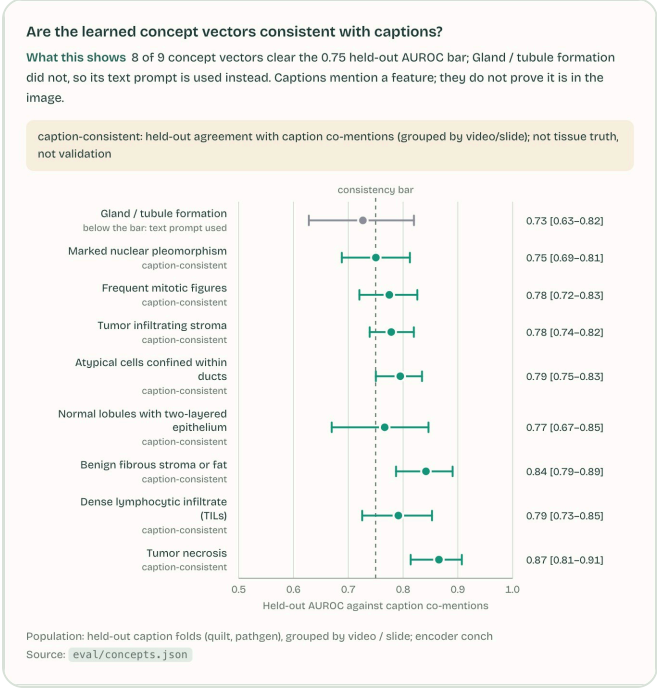
7.3b How well each feature alone separates invasive images. Above 0.5 means higher in invasive. Tumour infiltrating stroma is strongest (0.96).



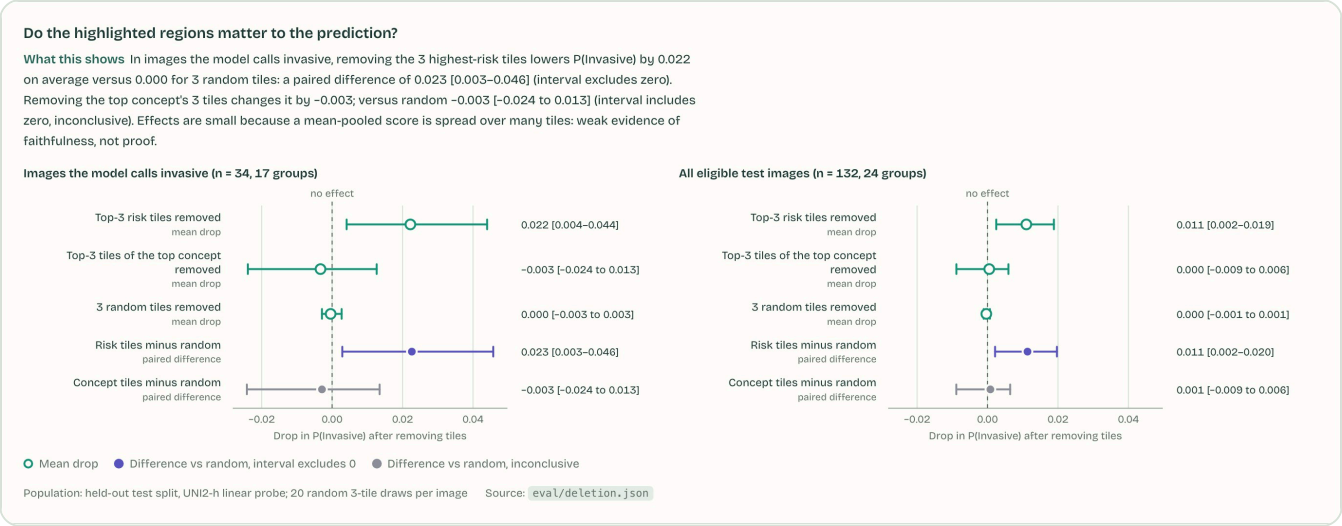
7.3c Spot check of the top-scoring regions. For each feature, 20 top regions were reviewed by an AI (Claude), not by a pathologist. Green bars are regions that looked like the named feature. Mitoses could not be resolved at review resolution; "normal lobules" mostly picked up fat and fibrous stroma.



7.3d Cross-organ sanity check. On colorectal tiles, text-prompt scores separate necrosis, lymphocytes and stroma well; the caption-trained lymphocyte vector is reversed there, and pleomorphism is weak.



7.3e Concept vectors against captions. 8 of 9 learned vectors clear the 0.75 bar on held-out caption data; gland/tubule formation does not, so its text prompt is used instead.



7.3f The deletion test: do the highlighted regions matter? Removing the three highest-risk regions lowers P(invasive) more than removing three random regions (difference 0.023, interval excludes zero). So the *heatmap* passes. Removing the regions picked by the *feature* attributions does not beat random, so the feature explanations are useful descriptions, but not proven to drive the read.

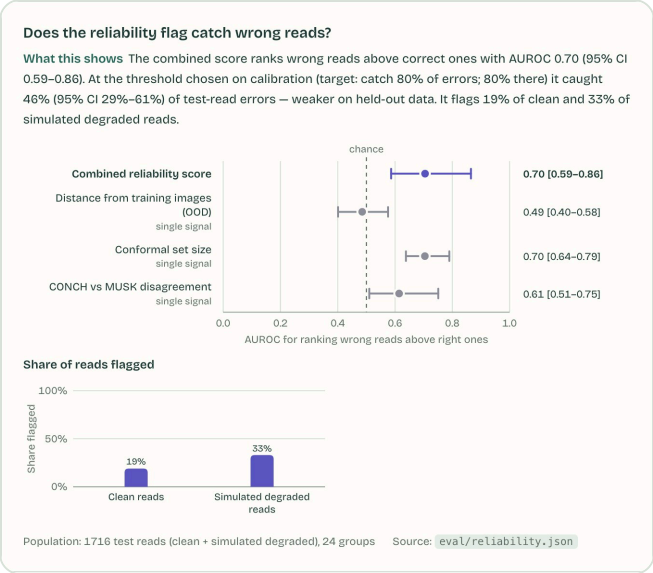
7.4 Frozen-section reliability (simulated)

Frozen sections are messier than permanent sections. To see how the read copes, test images were degraded on purpose. Four-class accuracy drops from 84% on clean images to about 67–73% at the worst setting, with JPEG compression hurting most. The reliability flag is meant to catch those errors. It ranks wrong reads above

right ones fairly well (AUROC 0.70) but at its chosen threshold it catches only 46% of errors, so it is a helpful warning light, not a safety net.



7.4a Accuracy against artifact severity for blur, JPEG compression, stain shift and resampling, with 95% intervals.



7.4b Does the flag catch wrong reads? The combined score beats each signal alone. It flags 19% of clean reads and 33% of degraded ones.



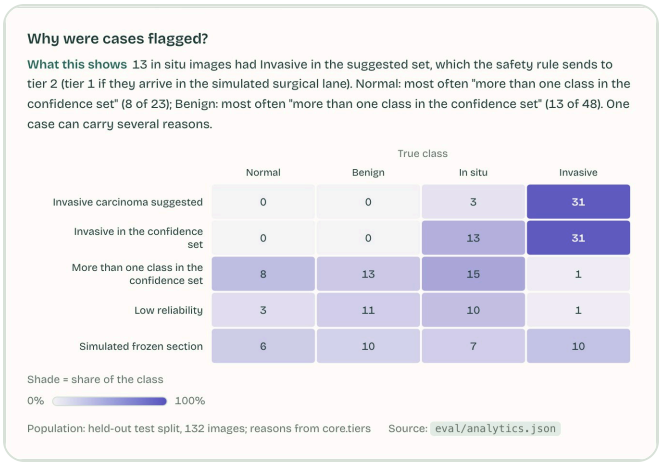
7.4c The simulated margin read. It never called tumour at a benign margin, and kept every truly tumour-free composite margin clear. But it found only 45% of involved margins on annotated slide windows: it is conservative.

7.5 Workflow: does prioritising actually help?

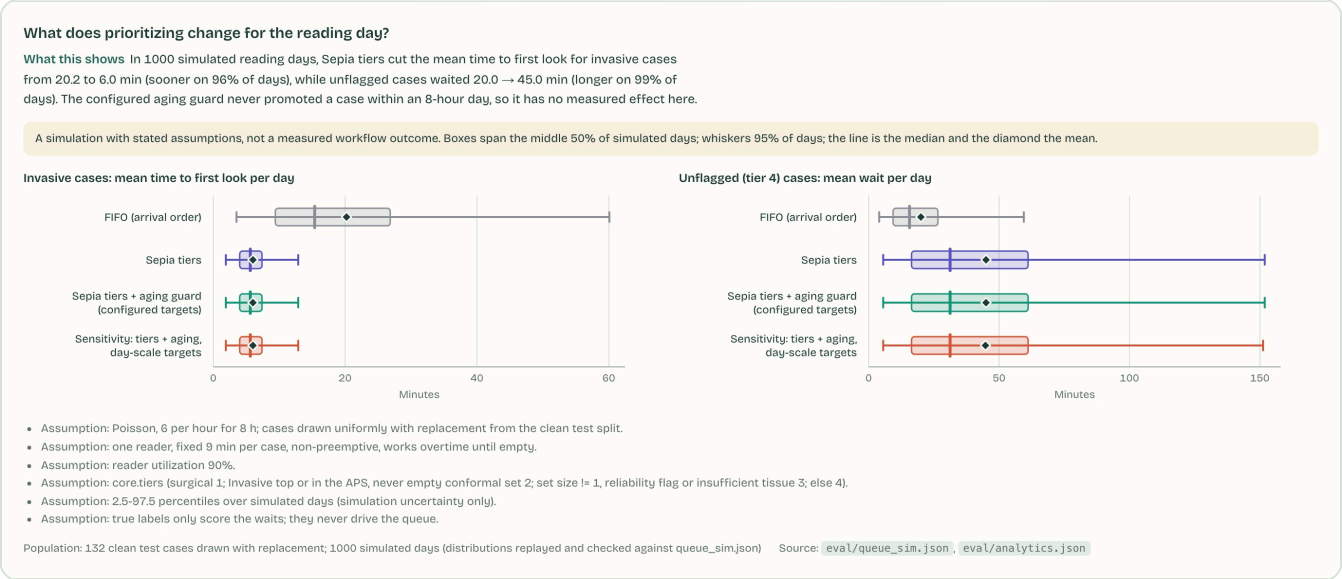
This is the clinical point of the app. All 31 invasive test images landed in tier 1 or 2, so none would have waited at the back of the queue. The price is that 9 non-invasive images were also moved up. In a simulation of 1,000 reading days (6 cases an hour for 8 hours, one reader, 9 minutes per case), Sepia's tiers cut the mean time before an invasive case is first opened from **20.2 to 6.0 minutes**. The cost is real and is shown next to the benefit: unflagged cases waited **45 minutes instead of 20** on average.



7.5a Tier by true class. Every invasive image is in tier 1 or 2.



7.5b Why cases were flagged. 13 in situ images had invasive in their set, which the safety rule sends up.

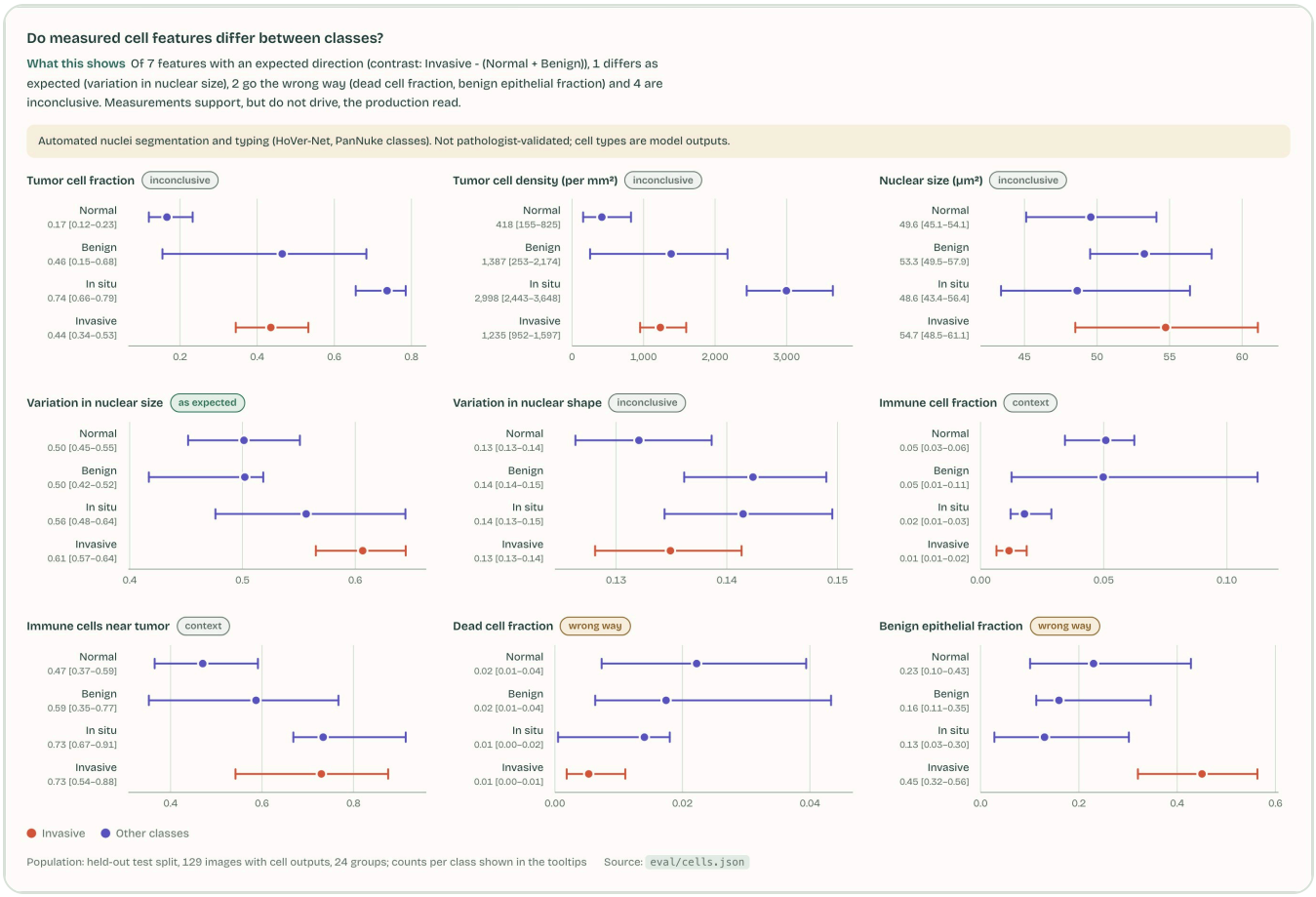


7.5c The simulated reading day. Left: time to first look for invasive cases under arrival order (FIFO) versus Sepia's tiers. Right: the extra wait imposed on unflagged cases. This is a model with stated assumptions, not a measured hospital outcome.

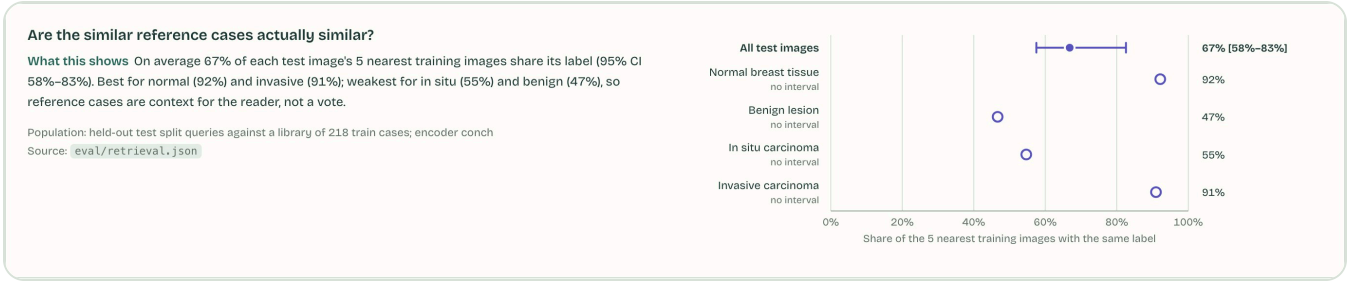
7.6 Cell measurements and similar cases

Counting and measuring nuclei was meant to add hard, physical evidence. Results were mixed: variation in nuclear size rose in invasive tissue as expected, but most other cell features were inconclusive and two went the wrong way. A model built on cell features alone performed poorly (macro-F1 0.33), so cells are shown as

context, not used for the read. Similar-case search returns a same-diagnosis neighbour 67% of the time on average: strong for normal and invasive tissue, weaker for benign and in situ.

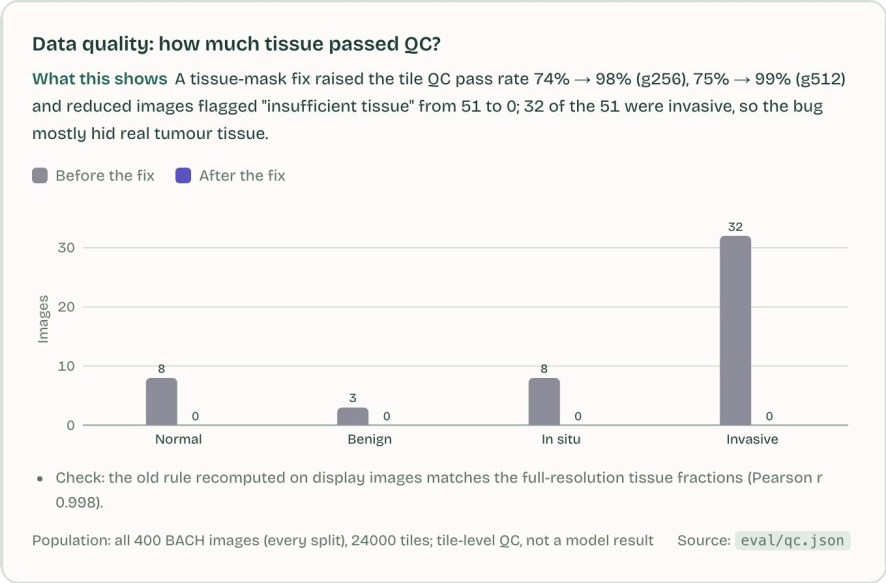


7.6a Cell features by class, with 95% intervals. Only variation in nuclear size differs in the expected direction with an interval that excludes zero.



7.6b Similar-case precision. Share of each test image's 5 nearest training images with the same diagnosis, by class.

7.7 Data quality



7.7 The tissue-mask fix. The original mask rejected pale H&E tissue, so 51 images were wrongly flagged "insufficient tissue", 32 of them invasive. The fix reduced that to 0 and raised the tile pass rate from 75% to 99%. All downstream steps and the evaluation were re-run afterwards.

8 · OUTCOMES

Outcomes, costs and limitations

What the project shows, what it costs, and what it does not claim.

What worked

- **Invasive cancer is found reliably on this data.** AUROC 0.998, 31 of 31 invasive test images caught, 97% specificity.
- **Confidence is honest enough to act on.** Calibration error halved, and confidence sets met their 90% target overall and covered every invasive image.
- **The safety rule does its job.** No invasive image would have waited in the bottom tiers, and the simulated wait before first look fell from about 20 minutes to 6.
- **Performance matches the published benchmark** for the same model on the same split.
- **The region heatmap is real evidence.** It passes the deletion test: removing highlighted regions changes the prediction more than removing random ones.
- **The pathologist stays in charge.** Orders are drafted, never placed; sign-out requires a decision; every action is logged.

What it costs

- Moving urgent cases forward pushes others back. In the simulation, unflagged cases waited about 25 minutes longer on average.
- The safety rule over-flags: 9 non-invasive test images were sent to tier 2, mostly in situ carcinoma.

What did not work, or is not proven

- **Some feature explanations are weak.** "Mitoses" and "normal lobules" did not point at the right tissue in the spot check, and the feature attributions did not pass the deletion test.
- **The reliability flag catches under half of the errors** at its chosen threshold.
- **The margin read is conservative.** It rarely raises a false alarm but misses about half of involved margins.
- **Cell measurements added little** to the read.
- **The conformal method was switched after viewing test results**, so its coverage figure is optimistic.

Limitations

- A research prototype on a small public dataset: 132 test images from 24 patient groups, so intervals are wide.
- BACH labels are diagnostic categories, not Nottingham grades. "Likely invasive" is a triage suggestion, not a diagnosis.
- The surgical lane is simulated: image borders stand in for inked margins, frozen-section artifacts are synthetic, and composite fields are built from test images.
- Feature scores are associations, not causes. Caption consistency measures how captions co-mention a feature, not tissue truth.
- Spot checks were done by an AI, not a pathologist.

- The queue result is a simulation with stated assumptions, not a measured workflow outcome.
- Stain rules are illustrative and require clinical review. No clinical validation and no hospital integration is claimed.

The one-sentence takeaway

Pathology foundation models can be turned into a triage tool that is fast, calibrated and open about its uncertainty; making every *explanation* as trustworthy as the prediction itself is the part that still needs work and expert review.

9 · CREDITS

Datasets, models and methods

BACH	Aresta G, et al. BACH: Grand challenge on breast cancer histology images. *Medical Image Analysis*, 2019. CC BY-NC-ND 4.0.
UNI2-h, CONCH	Mahmood Lab. Chen RJ, et al., *Nature Medicine* 2024 (UNI); Lu MY, et al., *Nature Medicine* 2024 (CONCH).
MUSK	Xiang J, et al. A vision–language foundation model for precision oncology. *Nature*, 2025.
H-optimus-0, PathGen-CLIP	Bioptimus; Sun Y, et al. PathGen-1.6M.
THUNDER	Split and published benchmark values, checked against the authors' leaderboard.
Quilt-1M, PathGen, NCT-CRC-HE-100K	Concept example images and captions; cross-organ sanity check (Kather JN, et al., 2019).
HoVer-Net / PanNuke, TIAToolbox	Nucleus segmentation and classification.
Conformal prediction	APS: Romano, Sesia & Candès, NeurIPS 2020. LAC: Sadinle, Lei & Wasserman, JASA 2019.
Compute and tooling	NCSA Delta, NCSA Lumen (gemma-4-31b-it summaries), and the MIT-licensed `ncsa/ashby-hackathon-2026` tooling.
Frontend	Original worklist by teammate adas286, extended into the Sepia workspace.

All figures in this document are screenshots of the running Sepia app (Model evaluation page and case views) taken on October 18, 2026, from the pipeline outputs in `artifacts/`. Numbers are quoted from `docs/RESULTS.md` and `artifacts/eval/*.json`.